

LOOPLEARN HACKATHON 2026

HealthBridge

AI-Powered Community Health
Information & Support Platform

*Making reliable health information and
community health resources easier to
access.*



THE PROBLEM

The Health Information Gap

Citizens struggle to find reliable health information, relevant educational content, and community resources that fit their needs.



THE OPPORTUNITY

HealthBridge combines verified resources, educational content, AI assistance, and intelligent search into one trusted platform.

OUR SOLUTION

One Platform. Three Connected Experiences.



INTELLIGENCE

What Makes HealthBridge Intelligent?

AI Health Education Assistant

Powered by Google Gemini API + RAG for grounded, trustworthy answers.

Retrieval-Augmented Generation

TF-IDF retrieval surfaces the most relevant verified content.

Health Resource Agent

Uses multiple search tools to find resources and articles.

THE AI FLOW

User Question

Intent Detection

Tool Selection

Parallel Tool Execution

Relevant Resources & Articles

Structured Response

EXAMPLE USER JOURNEY

From Question to Actionable Information

"What community resources help manage diabetes?"

A citizen asks a real, everyday health question.

THE AI PIPELINE

- 1 Intent detection understands the question
- 2 RAG retrieval finds verified knowledge
- 3 Resource Agent searches matching services
- 4 Gemini generates a clear, grounded answer

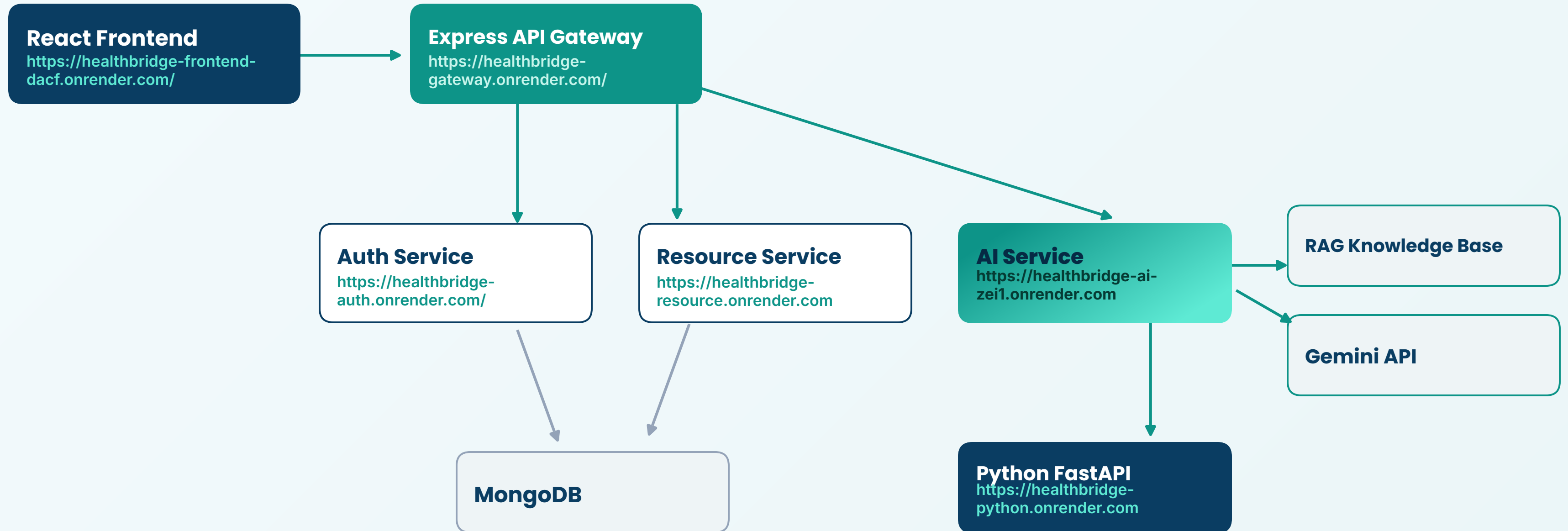
AI RESPONSE

Here are verified community programs, educational articles, and support services for diabetes care near you — with links to trusted resources on HealthBridge.

⚠️ **Safety disclaimer: educational information only — not medical advice.**

SYSTEM ARCHITECTURE

Modular, AI-Powered by Design



TECHNOLOGY STACK

Built on a Modern, Scalable Stack

Frontend

React 19 · Vite 8 · Tailwind CSS 4 · JSX · Bun

Backend

Node.js · Express.js · Microservices architecture

Data

MongoDB · Mongoose ODM

AI

Google Gemini API · RAG · TF-IDF retrieval · Health Resource Agent

Python

Python 3.11+ · FastAPI · Pydantic

Infrastructure

Docker · Docker Compose · Kubernetes · Terraform · Git/GitHub

SECURITY & TRUST

Responsible Engineering for Health

SECURITY FLOW



BUILT-IN SECURITY LAYERS

- bcrypt password hashing
- JSON Web Tokens (JWT)
- HTTP-only cookies
- Role-based authorization
- CORS & Helmet headers
- API rate limiting
- Input validation
- Secrets management
- AI health disclaimers

Aligned with Global Goals

03

Good Health & Well-Being

Reliable, verified health information and education for everyday decisions and healthier lives.

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Reduced Inequalities

Free, AI-assisted access to trusted health resources for underserved communities.

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Sustainable Communities

Connecting citizens with local organizations and community health services.

WHY WE'RE DIFFERENT

Beyond Search — Intelligent Support

Traditional Health Search

- Scattered, unverified results
- Generic keyword matching
- No personalized guidance
- Information overload
- No connection to real services
- Uncertain reliability

HealthBridge Intelligent Support

- AI Health Education Assistant
- RAG knowledge retrieval
- Intelligent Resource Agent
- Multi-user ecosystem
- Modular microservices
- Python AI backend
- Security-focused by design

THE TEAM

Thank You

Building HealthBridge to make trusted health information accessible to everyone.

HealthBridge · LoopLearn
Hackathon 2026

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